∴ Volume to be removed

= (Volume of the circle – Volume of the square)

$= \left\{\left[\pi\left(\sqrt{2}\right)^2 \times 1\right] - (2 \times 2 \times 1)\right\}$

$= 2\pi - 4$

146. One local and another express train were proceeding in the same direction on parallel tracks at 29 km/hour and 65 km/hour respectively. The driver of the former noticed that it took exactly 16 seconds for the faster train to pass by him. What is the length of the faster train?

(a) 60 m (b) 120 m

(c) 160 m (d) 240 m

Ans. (c) : Relative speed of express train to local train = 65 – 29 = 36 km/h

= 36 × (5/18) m/s = 10 m/s

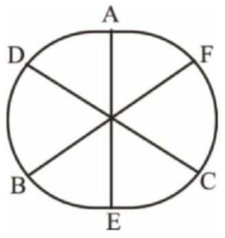
∴ Length of faster train = 10 × 16 = 160 m

147. A, B, C, D, E and F, not necessarily in that order are sitting on six chairs regularly placed around a round table. It is observed that A is between D and F, C is opposite D, and D and E are not on neighbouring chairs. Which one of the following pairs must be sitting on neighbouring chairs.

(a) A and B (b) C and E

(c) B and F (d) A and C

Ans. (b) : Based on the information given in the question following sitting arrangement can be made–



So C and E are sitting on the neighbouring chairs.

148. If in a certain code SAND is VDQG and BIRD is ELUG, then what is the code for MOVE?

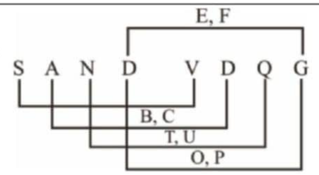
(a) PRYH

(b) ORTG

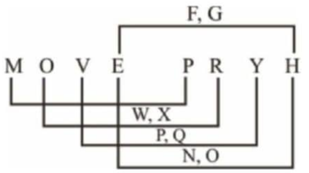
(c) NPUH

(d) ORYH

Ans. (a) :



Similarly



149. In a family, a couple has a son and a daughter. The age of the father is three times that of his daughter and the age of the son is half of this mother. The wife is nine years younger to their husband and the brother is seven years older than his sister. What is the age of the mother?

(a) 40 years (b) 45 years

(c) 50 years (d) 60 years

Ans. (d) : Let the daughter's age be x

Then father age = $3 \times x = 3x$

Mothers age = $3x - 9$

Son's age = $x + 7$

So $x + 7 = \frac{3x-9}{2} \Rightarrow 2x + 14 = 3x - 9$

$\Rightarrow x = 23$

So mothers age = 23 × 3 – 9 = 60 years

150. The missing fraction in the series given below is:

$$\frac{4}{9}, \frac{9}{20}, \ldots\ldots \frac{39}{86}$$

(a) $\frac{17}{40}$ (b) $\frac{19}{42}$

(c) $\frac{20}{45}$ (d) $\frac{29}{53}$

Ans. (b) : The pattern in the series is:

$\frac{4\times2+1}{9\times2+1} = \frac{9}{20}$

$\frac{9\times2+1}{20\times2+2} = \frac{19}{42}$

$\frac{19\times2+1}{42\times2+1} = \frac{39}{86}$

Hence the missing fraction would be $\frac{19}{42}$

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 1997

GENERAL STUDIES : PAPER-I

Time: 2 hours **Maximum Number: 200**

HISTORY

1. Match List-I with List-II and select the correct answer:

List-I	List-II
A. Gupta	1. Badami
B. Chandella	2. Panamalai
C. Chalukya	3. Khajuraho
D. Pallava	4. Deogarh

Codes:

	A	B	C	D
(a)	4	3	1	2
(b)	4	2	3	1
(c)	2	3	4	1
(d)	3	4	1	2

Ans. (a) : The Gupta had their capital at Deogarh, Chandella had their capital at Khajuraho, Chalukya's at Badami and Pallava's capital was at Panamalai.

2. Which one of the following statements regarding Ashoka stone pillars is incorrect?

(a) These are highly polished

(b) These are monolithic

(c) The shaft of pillars is tapering in shape

(d) These are parts of architectural structures

Ans. (d) : The number of these pillar articles is seven, which have been found in different places in the form of independent pillars. These stone pillars are well polished. Delhi-Topra is important in these pillar articles, initially it was buried in Saharanpur district of Uttar Pradesh. Brought to Delhi by Firoz Shah Tughlaq in the Middle Ages. Seven inscriptions of Ashoka are engraved on it. Other pillar inscriptions of Ashoka are Delhi, Meerut, Lauria-Arraj, Lauria Nandangarh, Rampurwa, Prayag. The stone pillars on which the state proclamations of Ashoka are engraved are generally called small pillar inscriptions.

3. Which one of the following pairs of composers in different languages and their works on the Mahabharata theme is correctly matched?

(a) Sarladasa – Bengali

(b) Kasirama – Oriya

(c) Tikkana – Marathi

(d) Pampa – Kannada

Ans. (d) : Pampa was a Kannada poet whose works reflected his philosophical beliefs. Vikramarjuna Vijaya, also known as Pampa Bharata, is a Kannada version of the Mahabharata of Vyasa.

4. Which one of the following scripts of ancient India was written from right to left?

(a) Brahmi (b) Nandagari

(c) Sharada (d) Kharoshti

Ans. (d) : Most of the inscriptions of Ashoka in ancient India are in Brahmi script. This script was written from left to right. Some of his inscriptions have also been found in Kharoshthi script which was written from right to left.

5. The famous dialogue between Nachiketa and Yama is mentioned in the:

(a) Chhandogyopanishad

(b) Mundakopanishad

(c) Kathopanishad

(d) Kenopanishad

Ans. (c) : Kathopanishad is the conversation between Nachiketa and Yamraj (God of death). Nachiketa was a seeker, and Yamraj was a Guru (knower of secrets). The Katha Upanishad is one of the mukhya Upanishads, embedded in the last short eight sections of the Kaṭha school of the Krishna Yajurveda. It is also known as Kāṭhaka Upanishad, and is listed as number 3 in the Muktika canon of 108 Upanishads

6. Milindapanha is in the form of a dialogue between the king Menander and the Buddhist monk:

(a) Nagasena

(b) Nagarjuna

(c) Nagabhatta

(d) Kumarilabhatta

Ans. (a) : Milindpanha is a famous Buddhist scripture written in Pali language, in which the question-answer between the Indo-Greek king Menander or Milanda and the Buddhist teacher Nagasena is compiled. It was composed in the first century AD.

7. Which one of the following edicts mentions the personal name of Asoka?

(a) Kalsi (b) Rummindei

(c) Special Kalinga Edict (d) Maski